

Predictors of Statistics Anxiety Among University Students: An Applied Regression Analysis

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Abstract

Statistics anxiety is a common academic challenge that may affect students' performance, confidence, and willingness to engage with statistical learning. This study examines factors associated with statistics anxiety among university students using a dataset of 447 participants. The dataset includes validated psychological measures related to statistics anxiety, math anxiety, anxiety sensitivity, perfectionism, and self-efficacy. Using applied regression methods, this project investigates several related research questions, including whether statistics anxiety differs across academic faculties, whether anxiety sensitivity improves the prediction of test anxiety beyond math anxiety, whether a linear model adequately describes the relationship between perfectionism and interpretation anxiety, whether prior statistics experience moderates the relationship between math anxiety and statistics anxiety, and which predictors form the most parsimonious model for help-seeking anxiety. The analysis applies regression-based ANOVA, partial F-tests, lack-of-fit testing, interaction modeling, model selection, and diagnostic checking. Overall, this study aims to identify important psychological and academic predictors of statistics anxiety and provide evidence that may support more targeted academic interventions.

Keywords: statistics anxiety, math anxiety, anxiety sensitivity, perfectionism, self-efficacy, multiple linear regression, model selection

1 Introduction and Background

Statistics anxiety is a multidimensional form of academic anxiety involving worry, negative thoughts, and discomfort when students engage with statistical tasks. Because statistics courses are required in many academic programs, this anxiety may affect students' confidence, course performance, willingness to ask for help, and future engagement with quantitative reasoning. Understanding predictors of statistics anxiety can therefore provide useful evidence for designing more targeted academic support. Although statistics anxiety is also closely related to math anxiety, the two constructs are not identical. Math anxiety reflects anxiety in mathematical situations more broadly, whereas statistics anxiety may also involve test anxiety, interpretation anxiety, asking-for-help anxiety, attitudes toward statistics instructors, and computational self-concept. In addition to math anxiety, broader psychological factors such as anxiety sensitivity, perfectionism, and self-efficacy may help explain why some students experience stronger statistics anxiety than others [1, 2, 3, 5, 6].

In this project we use the dataset *The Association of Self-Efficacy, Anxiety Sensitivity, and Perfectionism with Statistics and Math Anxiety* by McCaughey et al. [4]. The goal is to apply regression methods to

examine how academic background and psychological variables are associated with different dimensions of statistics anxiety among university students. Specifically, we study whether statistics test anxiety differs across academic faculties, whether anxiety sensitivity improves prediction beyond math anxiety, whether a linear model adequately describes the relationship between perfectionism and interpretation anxiety, whether prior statistics experience moderates the association between math anxiety and statistics anxiety, and which predictors form the most parsimonious model for asking-for-help anxiety.

Together, these research questions aim to examine whether broad academic grouping variables or more proximal psychological variables better explain individual differences in statistics anxiety. We connect regression modeling with a meaningful educational problem. By exploring psychological and academic predictors of statistics anxiety, the analysis may provide empirical evidence for designing more targeted academic support and intervention strategies for students who experience anxiety in statistics courses.

2 Data and Variables

This project uses the dataset *The Association of Self-Efficacy, Anxiety Sensitivity, and Perfectionism with Statistics and Math Anxiety* by McCaughey et al. [4]. The dataset contains 453 student responses and 97 variables, including demographic information, individual questionnaire items, and composite psychological scale scores. The original study reported a final analyzed sample of 447 students after exclusions; in our analyses, the sample size may vary slightly across research questions because complete-case analysis was used for the variables required in each model. Because the data are observational and cross-sectional, the regression models are used to explain and predict associations rather than to make causal claims.

The sample included students from Arts, Science, and Other academic faculties, with Science students forming the largest faculty group. Students also varied in whether they had prior statistics course experience. These variables are useful academic background predictors, but because students were not randomly assigned to faculties or prior-experience groups, they are interpreted as descriptive characteristics rather than causal factors. The demographic and academic background distributions are provided in the separate **appendix file**.

The dataset includes several validated psychological instruments. Statistics anxiety was measured using the Statistics Anxiety Rating Scale (STARS), which includes subscales for test and class anxiety, interpretation anxiety, asking-for-help anxiety, worth of statistics, fear of statistics teachers, and computational self-concept [1]. Math anxiety was measured using the Abbreviated Math Anxiety Scale (AMAS) [3]. Anxiety sensitivity was measured using the Anxiety Sensitivity Index-3 (ASI-3), which includes physical, cognitive, and social concern subscales [6]. Perfectionism-related predictors were measured using perfectionism scales including the Frost Multidimensional Perfectionism Scale, and self-efficacy was measured using a general self-efficacy scale [2, 5]. Table 1 lists the primary variables used in the regression analyses.

Although the original questionnaire items are ordinal Likert-type responses, the response variables used in the regression analyses are multi-item composite scale scores rather than single items. This distinction is important for the continuous-response requirement of linear regression. A single 1–5 Likert item should not be treated as truly continuous, but composite scores such as `stat.anx.tc`, `stat.anx.i`, `stat.anx.ah`,

and `math.anx` take on more possible values and are intended to measure underlying psychological constructs. Following common practice in applied psychological and educational research, these composite scale scores are therefore treated as approximately continuous outcomes or predictors. Because the composite outcomes are still bounded scale scores, this modeling choice is checked through residual-versus-fitted plots, Q–Q plots, variance diagnostics, VIF values, and influence diagnostics for the fitted models.

Missingness was limited for the composite variables used in the main regression analyses. The largest amount of missingness occurred in `perf.cog`, while most other composite variables had very few missing observations. Because missingness was minimal, each regression model was fit using complete cases for the variables required in that specific analysis. Additional sample-description plots, including the missing-data summary, age distribution, top reported majors, item-level response distributions, and demographic composition, are provided in the separate **appendix file**.

Table 1: Key Variables Used in the Regression Analyses

Role	Variable(s)	Use in Analysis
Statistics anxiety outcomes	<code>stat.anx.tc</code> , <code>stat.anx.i</code> , <code>stat.anx.ah</code>	Responses for test anxiety, interpretation anxiety, and asking-for-help anxiety.
Additional STARS subscale	<code>stat.anx.sc</code>	Computational self-concept; used descriptively.
Math anxiety	<code>math.anx</code>	Main predictor for test anxiety, moderation analysis, and model selection.
Anxiety sensitivity	<code>asi.phys</code> , <code>asi.cog</code> , <code>asi.social</code>	Added to assess incremental predictive value beyond math anxiety.
Perfectionism	<code>frost.com</code>	Predictor for the lack-of-fit analysis with interpretation anxiety.
Self-efficacy	<code>self.efficacy</code>	Candidate predictor in asking-for-help model selection.
Academic background	Faculty, prior statistics experience	Categorical predictors for dummy-variable regression and interaction modeling.

3 Exploratory Data Analysis

Before fitting the regression models, we examined the distributions, normal Q–Q patterns, correlations, and group-based patterns among the main composite variables. Because the original survey responses are bounded Likert-type items and the analysis uses composite scale scores, the EDA focused on whether the scale scores showed strong skewness, tail deviations, unusual outliers, or relationships that would affect later regression modeling.

Figure 1 summarizes three key exploratory checks: the distributions of the main composite scores, Q–Q plots of those scores, and their Pearson correlation structure. The distributions show that several composite variables are approximately unimodal, but some variables display noticeable skewness. For example, worth-of-statistics anxiety (`stat.anx.ws`) and computational self-concept (`stat.anx.sc`) are right-skewed, while self-efficacy and personal standards show more left-skewed patterns. The Q–Q plots show that many variables are approximately linear in the center but deviate in the tails, which is expected for bounded psychological scale scores. Therefore, normality is evaluated mainly through residual diagnostics after model fitting rather than through the raw composite variables alone.

The Pearson correlation heatmap supports the main modeling choices. Math anxiety is strongly and positively associated with statistics test anxiety and interpretation anxiety, motivating its use as a central predictor in several models. The ASI subscales are also highly correlated with the ASI total score, which is expected because the subscales are components of the total anxiety sensitivity measure. Therefore, models using `asi.phys`, `asi.cog`, and `asi.social` do not also include the ASI total score, avoiding construction-

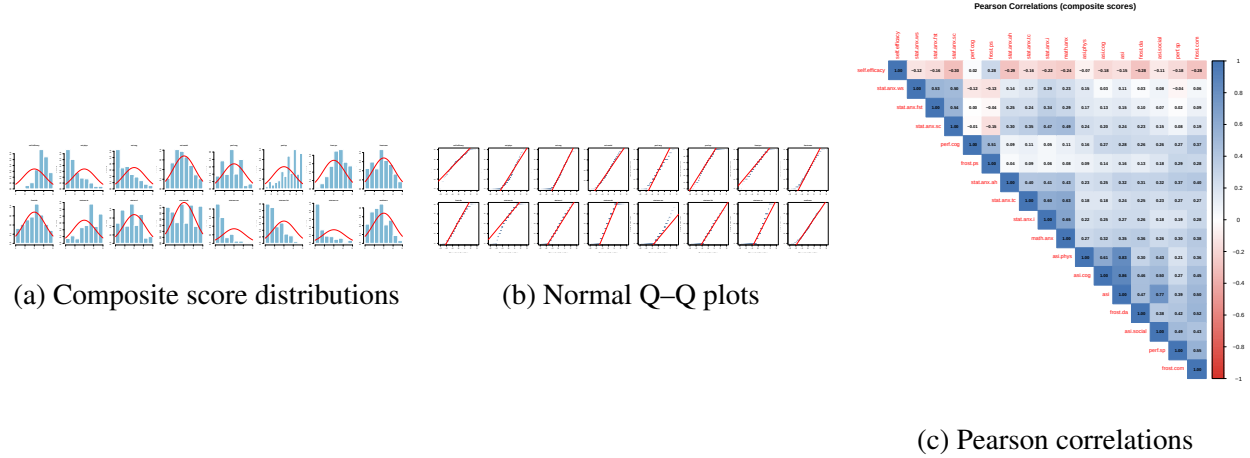


Figure 1: Exploratory summaries of the main composite variables.

based multicollinearity. Self-efficacy shows negative correlations with several anxiety-related outcomes, suggesting that students with greater confidence in their own abilities tend to report lower anxiety. These patterns support the use of regression models to examine the unique contribution of each predictor while also motivating later diagnostic checks for multicollinearity.

To examine preliminary group differences, Figure 2 shows boxplots of selected anxiety-related variables by academic faculty and prior statistics experience. These plots directly motivate the later dummy-variable regression and moderation analyses. The faculty-based boxplots suggest that Arts students tend to report somewhat higher statistics test anxiety than Science and Other students, although the distributions overlap substantially. This supports formally testing faculty differences while also suggesting that faculty alone is unlikely to explain a large portion of individual variation in anxiety.

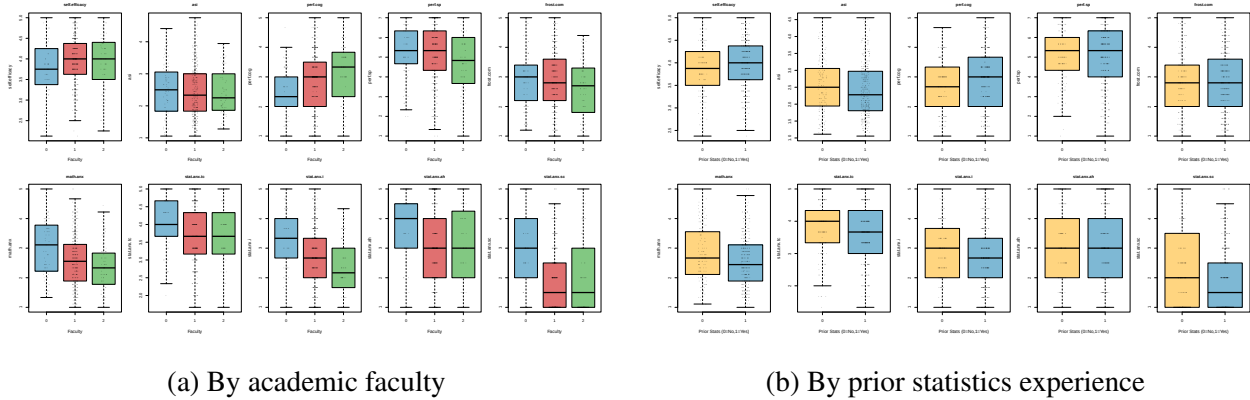


Figure 2: Group-based exploratory comparisons for academic background variables.

The prior-statistics-experience boxplots suggest that students without prior statistics experience tend to report higher math anxiety and statistics test anxiety than students with prior experience. This pattern motivates the interaction analysis testing whether prior statistics experience changes the association between math anxiety and statistics anxiety. Overall, the EDA supports the main modeling strategy: using math

anxiety as a key continuous predictor, treating faculty and prior statistics experience as categorical background variables, checking ASI dimensions for incremental predictive value, and evaluating assumptions through model-specific residual diagnostics. Additional exploratory plots, including Spearman correlations, outlier summaries, gender-based boxplots, missing-data summaries, and item-level response distributions, are provided in the **separate appendix file**.

4 Regression Analyses and Results

All analyses were conducted using the multiple linear regression framework,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \epsilon_i,$$

where Y_i is an anxiety-related composite response for student i , the X 's are explanatory variables, and ϵ_i is the random error term. Depending on the research question, this framework was used for dummy-variable regression, nested-model partial F -testing, lack-of-fit testing, interaction modeling, and model selection. For each research question, we state the motivation, model, hypotheses, results, diagnostic checks, and conclusion together so that the inference is directly connected to the model being tested.

4.1 RQ1: Faculty Differences in Statistics Test Anxiety

Research question. Do mean levels of statistics test anxiety differ across academic faculties?

We first examined whether broad academic grouping was associated with statistics test anxiety before considering more proximal psychological predictors. Faculty was treated as a categorical background variable rather than a causal factor, with Arts as the reference group. The response variable was statistics test and class anxiety, $Y_i = \text{stat.anx.ttc}$. After excluding three observations with missing values on the response, the analytic sample was $N = 450$.

The dummy-variable regression model was

$$Y_i = \beta_0 + \beta_1 D_{\text{Science},i} + \beta_2 D_{\text{Other},i} + \epsilon_i,$$

where $D_{\text{Science},i} = 1$ for Science students and $D_{\text{Other},i} = 1$ for students in Other faculties. Under this coding, β_0 is the mean test anxiety for Arts students, while β_1 and β_2 are mean differences from Arts. The hypotheses were

$$H_0 : \beta_1 = \beta_2 = 0 \quad \text{vs.} \quad H_a : \text{at least one of } \beta_1, \beta_2 \neq 0.$$

The fitted model was

$$\hat{Y}_i = 3.978 - 0.345 D_{\text{Science},i} - 0.434 D_{\text{Other},i}.$$

Thus, the estimated mean statistics test anxiety was 3.978 for Arts students, 3.632 for Science students, and 3.544 for students in Other faculties. Table 2 and Figure 3 show the same pattern: Arts students had the highest average test anxiety, although the group distributions overlap substantially.

The overall regression-based ANOVA test was statistically significant, $F(2, 447) = 4.278, p = .014$, so we

Table 2: Statistics Test Anxiety by Academic Faculty

Faculty	<i>n</i>	Mean	SD
Arts	67	3.978	0.924
Science	315	3.632	0.948
Other	68	3.544	1.038

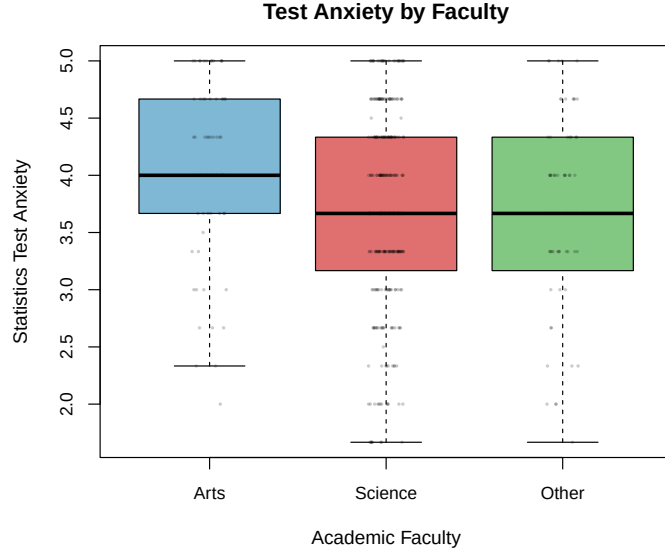


Figure 3: Statistics test anxiety by academic faculty.

reject H_0 and conclude that mean statistics test anxiety differs across at least one faculty group. However, the model explained only a small proportion of the variation in test anxiety, $R^2 = .019$, so the faculty effect is statistically detectable but practically modest.

The coefficient estimates show that, compared with Arts students, Science students reported significantly lower statistics test anxiety with $\hat{\beta}_1 = -0.345$, $t(447) = -2.677$, $p = .008$. Students in Other faculties also reported significantly lower test anxiety than Arts students with $\hat{\beta}_2 = -0.434$, $t(447) = -2.627$, $p = .009$. Tukey post-hoc comparisons supported the same interpretation: Arts differed from Science ($p = .021$) and Other ($p = .024$), while Science and Other did not differ significantly ($p = .771$).

Diagnostic checks supported the adequacy of this dummy-variable regression model. Levene’s test did not indicate unequal variances, $F(2, 447) = 0.539$, $p = .584$. The residual-versus-fitted plot showed the expected vertical bands for a categorical predictor model without a strong systematic pattern; the scale-location plot did not suggest severe heteroscedasticity; the Q–Q plot showed only mild tail deviation and discreteness, which is expected for a bounded composite scale but not severe enough to undermine the ANOVA interpretation; and the leverage plot did not suggest that the result was driven by a single influential observation. The full diagnostic panel is provided in the separate **appendix file**.

Overall, faculty membership was significantly associated with statistics test anxiety, with Arts students reporting higher average anxiety than both Science students and students in Other faculties. However, the small R^2 and substantial overlap in group distributions indicate that faculty explains only a limited share of individual differences, motivating the next analysis of psychological predictors.

4.2 RQ2: Incremental Validity of Anxiety Sensitivity

Research question. Does adding anxiety sensitivity dimensions improve prediction of statistics test anxiety beyond math anxiety alone?

We next examined whether anxiety sensitivity provides incremental predictive value after accounting for math anxiety, which is already expected to be strongly associated with statistics anxiety. The response variable was statistics test and class anxiety, $Y_i = \text{stat.anx.tcc}$. Complete-case analysis yielded $N = 449$. The reduced model included only math anxiety, while the full model added the three ASI-3 dimensions: physical concerns, cognitive concerns, and social concerns:

Reduced: $Y_i = \beta_0 + \beta_1(\text{math.anx})_i + \epsilon_i$,

Full: $Y_i = \beta_0 + \beta_1(\text{math.anx})_i + \beta_2(\text{asi.phys})_i + \beta_3(\text{asi.cog})_i + \beta_4(\text{asi.social})_i + \epsilon_i$.

The ASI total score was not included with the ASI subscales because the total score is constructed from the same components. The partial F -test hypotheses were

$$H_0 : \beta_2 = \beta_3 = \beta_4 = 0 \quad \text{vs.} \quad H_a : \text{at least one of } \beta_2, \beta_3, \beta_4 \neq 0.$$

Table 3 summarizes the nested-model comparison and the full-model coefficients. Adding the ASI dimensions decreased SSE from 254.25 to 249.88 and increased R^2 from 0.3914 to 0.4019, giving $\Delta R^2 = 0.0104$. Thus, the ASI dimensions explained only about 1.04% additional variation in statistics test anxiety beyond math anxiety alone. The partial F -test was $F(3, 444) = 2.5854$, $p = 0.05268$. Since $p > .05$, we fail to reject H_0 at the 0.05 level. Therefore, there is not sufficient evidence that the three ASI dimensions jointly improve prediction beyond math anxiety, although the result is borderline.

Table 3: RQ2 Model Comparison and Full-Model Coefficients

(a) Nested-model comparison						(b) Full-model coefficients					
Model	df	SSE	R^2	Adj. R^2	AIC	Predictor	Est.	SE	t	p	95% CI
Reduced	447	254.25	0.3914	0.3901	1024.85	Intercept	1.668	0.145	11.50	< .001	[1.38, 1.95]
Full	444	249.88	0.4019	0.3965	1023.08	math.anx	0.690	0.044	15.80	< .001	[0.604, 0.776]
						asi.phys	0.004	0.048	0.09	.929	[-0.090, 0.098]
						asi.cog	-0.073	0.047	-1.55	.122	[-0.165, 0.020]
						asi.social	0.121	0.045	2.67	.008	[0.032, 0.211]

The coefficient estimates show that math anxiety remained the dominant predictor. Holding the ASI dimensions constant, a one-unit increase in math anxiety was associated with an estimated 0.690-point increase in statistics test anxiety. Among the three ASI dimensions, only social concerns showed a statistically significant positive association with statistics test anxiety after adjusting for math anxiety and the other ASI dimensions.

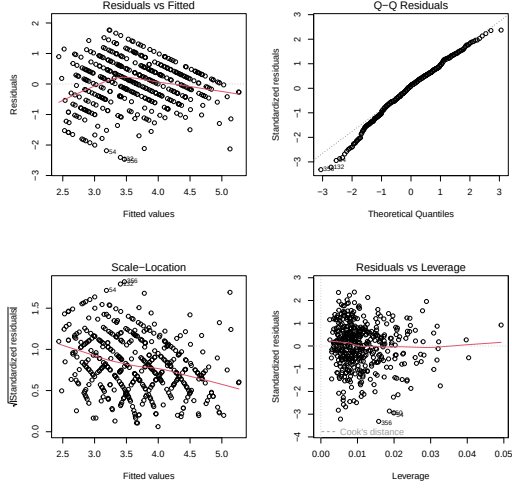


Figure 4: Diagnostic plots for the full RQ2 model.

Model diagnostics suggested some assumption concerns but did not change the substantive conclusion. As shown in Figure 4, the residual-versus-fitted and scale-location plots suggested mild curvature and non-constant residual spread, and the Breusch–Pagan test indicated heteroscedasticity, $BP = 26.887$, $p < .001$. The Q–Q plot was approximately linear in the center but showed some tail deviation, consistent with the significant Shapiro–Wilk test, $W = 0.9860$, $p < .001$. Given the large sample size and bounded composite outcome, this normality result was interpreted together with the visual diagnostics. VIF values were low for all predictors: 1.14 for `math.anx`, 1.66 for `asi.phys`, 1.83 for `asi.cog`, and 1.38 for `asi.social`, indicating no serious multicollinearity.

Because the Breusch–Pagan test indicated non-constant variance, HC3 robust standard errors were examined as a sensitivity check. The robust standard errors did not change the substantive conclusion: `math.anxiety` remained strongly significant, `asi.social` remained significant, and `asi.phys` and `asi.cog` remained non-significant. Influence diagnostics identified several potentially influential observations using the conservative $4/n$ Cook’s distance rule, but no single observation appeared to dominate the model. Additional diagnostic outputs, including Cook’s distance and added-variable plots, are provided in the separate **appendix file**.

Overall, the partial F -test suggests that the three ASI dimensions do not jointly add statistically significant predictive value beyond math anxiety at the 0.05 level. The practical improvement in model fit was small, with only a 1.04% increase in explained variance. The main conclusion is that math anxiety is the primary predictor of statistics test anxiety in this model, while social anxiety sensitivity may have a modest unique association.

4.3 RQ3: Lack-of-Fit Test for the Perfectionism Model

Research question. Does a first-order linear model adequately describe the relationship between perfectionism-related concern over mistakes and statistics interpretation anxiety?

This question evaluates whether the association between perfectionism-related concern over mistakes and interpretation anxiety can be represented by a simple linear trend. This is useful because, if the linear form is adequate, we can interpret the slope directly without adding unnecessary nonlinear or saturated terms. The response variable was interpretation anxiety, $Y_i = \text{stat.anx.i}$, and the predictor was concern over mistakes, $X_i = \text{frost.com}$. Complete-case analysis yielded $N = 451$.

The first-order linear model was

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i.$$

To test lack of fit, this model was compared with a saturated model that treated `frost.com` as a categorical

predictor, allowing a separate mean response at each observed predictor value. The hypotheses were

$$H_0 : E(Y | X) = \beta_0 + \beta_1 X \quad \text{vs.} \quad H_a : E(Y | X) \neq \beta_0 + \beta_1 X.$$

The fitted linear model showed a positive association between concern over mistakes and interpretation anxiety. The estimated slope was $\hat{\beta}_1 = 0.287$, with $SE = 0.0461$, $t = 6.22$, $p < .001$, and 95% CI $[0.196, 0.378]$. Thus, higher concern over mistakes was associated with higher statistics interpretation anxiety. The linear model explained a modest portion of the variation in interpretation anxiety, with $R^2 = 0.0793$ and adjusted $R^2 = 0.0772$.

Table 4: RQ3 Linear Model and Lack-of-Fit Results

(a) Linear-model coefficient						(b) Lack-of-fit comparison				
Predictor	Est.	SE	t	p	95% CI	Model	df	SSE	AIC	BIC
Intercept	1.920	0.139	13.80	< .001	[1.65, 2.19]	Linear	449	413.66	1246.91	1259.24
frost.com	0.287	0.046	6.22	< .001	[0.196, 0.378]	Saturated	427	387.97	1261.99	1364.77

The lack-of-fit test compared the linear model with the saturated model. The test gave $F(22, 427) = 1.2854$, $p = 0.1751$. Since $p > .05$, we fail to reject H_0 . Therefore, there is not sufficient evidence that the first-order linear model is inadequate. Although the saturated model reduced SSE from 413.66 to 387.97, the improvement was not statistically significant and came with many additional parameters.

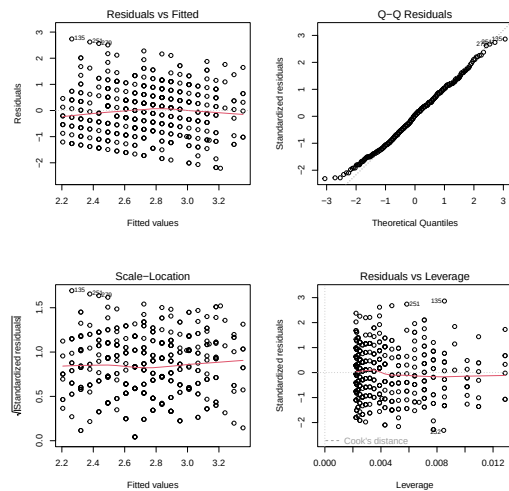


Figure 5: Diagnostic plots for the RQ3 linear model.

Model diagnostics supported the adequacy of the first-order linear model. As shown in Figure 5, the residual-versus-fitted plot showed no strong systematic curvature, and the scale-location plot did not suggest severe non-constant variance. This was supported by the Breusch–Pagan test, $BP = 0.0381$, $p = 0.8453$, indicating no evidence of heteroscedasticity. The Q–Q plot was approximately linear with mild tail deviation, and the Shapiro–Wilk test was borderline but not significant at the 0.05 level, $W = 0.9936$, $p = 0.0550$. Influence diagnostics flagged 43 observations using screening rules, but the maximum Cook’s distance was only 0.0353 and the largest absolute studentized residual was 2.887, so no single observation appeared to dominate the fitted model.

Overall, the lack-of-fit test suggests that a first-order linear model is adequate for describing the relationship between concern over mistakes and interpretation anxiety. Concern over mistakes is positively associated with interpretation anxiety, but the model explains a modest amount of variation. Therefore, the simple linear model is retained rather than adding unnecessary nonlinear or saturated terms.

4.4 RQ4: Moderation by Prior Statistics Experience

Research question. Does the association between math anxiety and statistics test anxiety depend on whether a student has prior statistics course experience?

This question tests whether prior statistics experience changes the strength of the relationship between math anxiety and statistics test anxiety. This is important because students with previous exposure to statistics may respond differently to math anxiety than students encountering statistics with less prior experience.

The response variable was statistics test and class anxiety, $Y_i = \text{stat.anx.tc}$. Math anxiety was centered to improve interpretation of the main effects, and prior statistics experience was treated as a categorical moderator. Complete-case analysis yielded $N = 449$, with 138 students reporting no prior statistics experience and 311 students reporting prior experience. The interaction model was

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 (X_i D_i) + \epsilon_i,$$

where $X_i = \text{math.anx}$ is centered math anxiety and $D_i = \text{stats.history}_i$ indicates prior statistics experience. The interaction coefficient β_3 tests whether the math-anxiety slope differs by prior statistics experience. The hypotheses were

$$H_0 : \beta_3 = 0 \quad \text{vs.} \quad H_a : \beta_3 \neq 0.$$

Table 5 summarizes the interaction model. The model explained a substantial proportion of the variation in statistics test anxiety, with $R^2 = 0.3996$ and adjusted $R^2 = 0.3956$. The main effect of centered math anxiety was positive and significant, $\hat{\beta}_1 = 0.545$, $p < .001$, indicating that among students without prior statistics experience, higher math anxiety was associated with higher statistics test anxiety. The main effect of prior statistics experience was not significant at average math anxiety, $\hat{\beta}_2 = -0.0553$, $p = .477$. Most importantly, the interaction term was positive and significant, $\hat{\beta}_3 = 0.215$, $SE = 0.0891$, $t = 2.42$, $p = .0160$, with 95% CI [0.0403, 0.390]. The equivalent one-degree-of-freedom interaction test gave $F(1, 445) = 5.8442$, $p = .0160$. Therefore, we reject H_0 and conclude that prior statistics experience significantly moderates the association between math anxiety and statistics test anxiety.

Table 5: RQ4 Interaction Model Results

(a) Coefficient estimates						(b) Model fit	
Predictor	Est.	SE	<i>t</i>	<i>p</i>	95% CI	Quantity	Value
Intercept	3.72	0.0649	57.3	< .001	[3.59, 3.84]	<i>N</i>	449
math_anx_c	0.545	0.0741	7.36	< .001	[0.399, 0.691]	Error df	445
stats.history1	-0.0553	0.0777	-0.711	.477	[-0.208, 0.097]	SSE	250.82
Interaction	0.215	0.0891	2.42	.016	[0.040, 0.390]	MSE	0.5636
						R^2	0.3996
						Adj. R^2	0.3956
						AIC	1022.77
						BIC	1043.30

Figure 6 visualizes the interaction. The fitted line for students with prior statistics experience is steeper than the fitted line for students without prior experience. This means that math anxiety is positively associated

with statistics test anxiety in both groups, but the association is stronger among students with prior statistics experience. In terms of slopes, the estimated math-anxiety slope for students without prior statistics experience is 0.545, while the estimated slope for students with prior statistics experience is $0.545 + 0.215 = 0.760$.

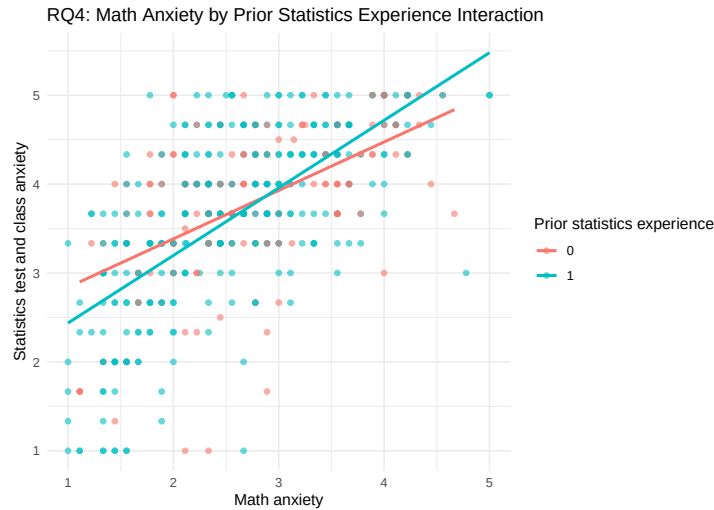


Figure 6: Interaction between math anxiety and prior statistics experience in predicting statistics test and class anxiety.

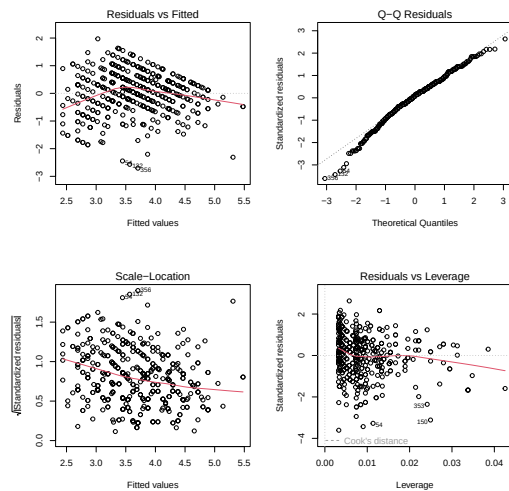


Figure 7: Diagnostic plots for the RQ4 interaction model.

Model diagnostics indicated some assumption concerns. As shown in Figure 7, the residual-versus-fitted and scale-location plots suggested non-constant residual spread, and the Breusch–Pagan test confirmed heteroscedasticity, $BP = 16.864$, $p = .000754$. The Q–Q plot showed tail deviation, and the Shapiro–Wilk test rejected residual normality, $W = 0.9833$, $p < .001$. VIF values were 3.28 for centered math anxiety, 1.02 for prior statistics experience, and 3.25 for the interaction term, so multicollinearity was not severe. Because heteroscedasticity was detected, HC3 robust standard errors were examined; the interaction remained significant, $\hat{\beta}_3 = 0.215$, robust $SE = 0.0914$, $p = .0190$. Influence diagnostics flagged 56 observations using screening rules, with maximum Cook’s distance 0.0620, maximum leverage 0.0425, and maximum absolute studentized residual 3.658. These cases should be interpreted as diagnostic flags rather than automatic exclusions, and no single observation was removed from the analysis.

Overall, prior statistics experience significantly moderated the relationship between math anxiety and statistics test anxiety. The positive interaction suggests that the math anxiety–statistics anxiety association was stronger among students with prior statistics experience. Although diagnostic checks showed heteroscedasticity and tail deviation, the HC3 robust standard error sensitivity check preserved the significance of the interaction, so the substantive conclusion remained unchanged.

4.5 RQ5: Model Selection for Asking-for-Help Anxiety

Research question. Which combination of self-efficacy, anxiety sensitivity, and math anxiety provides the most appropriate model for predicting statistics asking-for-help anxiety?

This question compares psychologically meaningful predictors to identify a model that balances predictive performance, parsimony, and interpretability. These psychologically meaningful predictors each represents a different possible source of asking-for-help anxiety: students' confidence in their own ability, broader sensitivity to anxiety symptoms, and anxiety about mathematical situations. Including all three allows us to evaluate whether help-seeking anxiety is best predicted by general self-beliefs, anxiety-related vulnerability, math-specific anxiety, or some combination of these factors. Because ordinary R^2 increases when predictors are added, model selection was based on adjusted R^2 , AIC, BIC, PRESS/LOOCV RMSE, 10-fold CV RMSE, and interpretability rather than R^2 alone.

The response was asking-for-help anxiety, $Y_i = \text{stat.anx.ah}$, and the candidate predictors were `self.efficacy`, `asi`, and `math.anx`. Complete-case analysis yielded $N = 451$. All $2^3 - 1 = 7$ non-empty subsets were compared. The full candidate model was

$$Y_i = \beta_0 + \beta_1(\text{self.efficacy})_i + \beta_2(\text{asi})_i + \beta_3(\text{math.anx})_i + \epsilon_i,$$

with reduced models formed by omitting one or more predictors. The intercept-only null model was included only as a baseline reference.

Selection rule. Smaller AIC, BIC, PRESS, LOOCV RMSE, and 10-fold CV RMSE indicate better models, while larger adjusted R^2 indicates better penalized explanatory fit.

Table 6: RQ5 Model Selection and Selected Model Coefficients

(a) Model selection and validation					(b) Selected model coefficients				
Model	Adj. R^2	AIC	BIC	10-fold RMSE	Predictor	Est.	SE	p	95% CI
SE_ASI_MA	0.2458	1361.99	1382.55	1.0916	Intercept	2.676	0.422	< .001	[1.85, 3.50]
SE_MA	0.2199	1376.25	1392.70	1.1087	self.efficacy	-0.361	0.0835	< .001	[-0.525, -0.197]
ASI_MA	0.2160	1378.48	1394.93	1.1128	asi	0.280	0.0692	< .001	[0.144, 0.416]
MA	0.1855	1394.71	1407.04	1.1322	math.anx	0.469	0.0639	< .001	[0.344, 0.595]
SE_ASI	0.1567	1411.35	1427.79	1.1542					
ASI	0.0993	1440.09	1452.42	1.1920					
SE	0.0812	1449.04	1461.38	1.2043					

Table 6(a) shows that the full three-predictor model, `SE_ASI_MA`, was selected by the main criteria considered: it had the lowest AIC, lowest BIC, highest adjusted R^2 , and lowest 10-fold CV RMSE. It also improved substantially over the intercept-only null model, which had $\text{AIC} = 1486.24$, $\text{BIC} = 1494.46$, and $\text{LOOCV RMSE} = 1.2543$. Although `SE_ASI_MA` is the most complex candidate model, it was retained because the criteria consistently favored it. The differences among the top models are not dramatic, especially among `SE_ASI_MA`, `SE_MA`, and `ASI_MA`, but the consistent advantage of the full model across both information criteria and validation metrics suggests that all three predictors contribute useful information.

Therefore, the selected model was

$$\widehat{\text{stat.anx.ah}} = \hat{\beta}_0 + \hat{\beta}_1(\text{self.efficacy}) + \hat{\beta}_2(\text{asi}) + \hat{\beta}_3(\text{math.anx}).$$

Table 6(b) shows the selected model coefficient estimates. The selected model had $R^2 = 0.2508$ and adjusted $R^2 = 0.2458$, explaining about 25.1% of the variation in asking-for-help anxiety. Holding the other predictors constant, self-efficacy was negatively associated with asking-for-help anxiety, while anxiety sensitivity and math anxiety were positively associated with asking-for-help anxiety. The VIF values were low: 1.067 for self-efficacy, 1.143 for anxiety sensitivity, and 1.186 for math anxiety, suggesting no serious multicollinearity.

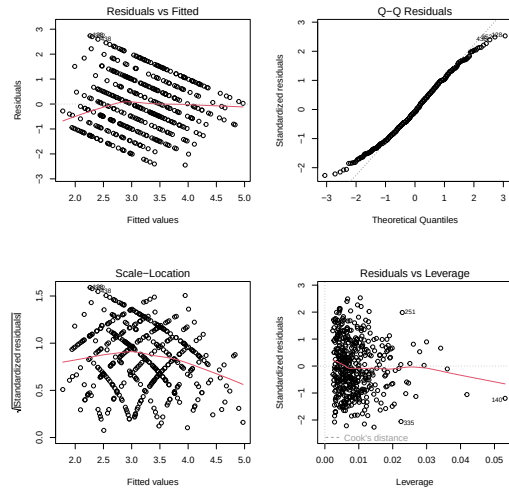


Figure 8: Diagnostic plots for the selected RQ5 model.

Model diagnostics showed some assumption concerns. As shown in Figure 8, the residual-versus-fitted and scale-location plots suggested some non-constant residual spread, and the Breusch–Pagan test supported heteroscedasticity, $BP = 11.471$, $p = .00943$. The Q–Q plot showed mild tail deviation, and the Shapiro–Wilk test rejected residual normality, $W = 0.9886$, $p = .00133$. Because heteroscedasticity was detected, HC3 robust standard errors were used as a sensitivity check; all three predictors remained statistically significant. Influence diagnostics flagged 47 observations using screening rules, but the maximum Cook’s distance was only 0.0243 and the largest absolute studentized residual was 2.544, so no single observation appeared to dominate the selected model.

4.5.1 Validation & Generalization

Validation was included to assess whether the selected model generalized beyond in-sample fit. The SE_{ASI_MA} model had the lowest PRESS, lowest LOOCV RMSE, and lowest 10-fold CV RMSE among all candidate models. Its 10-fold CV RMSE was 1.0916, compared with 1.1087 for the next-best SE_{MA} model and 1.1322 for the simpler math-anxiety-only model. Thus, the selected model was not chosen only because it fit the current sample well; it also had the best estimated out-of-sample predictive performance.

Generalization should still be interpreted cautiously. The data are observational and cross-sectional, so the selected model supports prediction and association rather than causal claims. The variables are bounded scale-based composites, which helps explain the visible banding in residual plots and the significant normality test. Overall, the full three-predictor model provides the best balance of fit, parsimony, and validation performance among the candidate models, but external validation in an independent student sample would be needed before making broader claims.

5 Discussion and Conclusions

Across the five research questions, the results suggest that statistics anxiety is related to both academic background variables and psychological predictors, but the psychological predictors generally explained more meaningful variation than broad group membership. Faculty differences were statistically significant

in RQ1, with Arts students reporting higher average statistics test anxiety than Science and Other students. However, the small R^2 indicated that faculty explained only a limited portion of individual differences. In contrast, math anxiety was consistently important across several models, especially in RQ2 and RQ4, where it was strongly associated with statistics test and class anxiety.

The partial F -test in RQ2 showed that the three anxiety sensitivity subscales did not jointly add statistically significant predictive value beyond math anxiety at the 0.05 level, although the result was borderline. This suggests that math anxiety already captures much of the variation in statistics test anxiety that is relevant for this model. However, social anxiety sensitivity remained individually significant in the full model, suggesting that concerns about socially visible anxiety symptoms may still have a modest unique association with statistics test anxiety. In RQ3, the lack-of-fit test supported the adequacy of a first-order linear model between perfectionism-related concern over mistakes and interpretation anxiety. This allowed us to retain the simpler linear model and interpret the positive slope directly.

RQ4 provided evidence of moderation: prior statistics experience significantly changed the strength of the association between math anxiety and statistics test anxiety. The positive interaction indicated that the math anxiety–statistics anxiety relationship was stronger among students with prior statistics experience. One possible interpretation is that students with prior exposure to statistics may have more concrete experiences with statistical tasks, so their math anxiety becomes more strongly connected to statistics-specific anxiety. However, because this is an observational study, this result should be interpreted as an association rather than evidence that prior statistics experience causes a stronger anxiety relationship.

RQ5 extended the analysis from inference to model selection and prediction. Among the seven candidate models for asking-for-help anxiety, the model including self-efficacy, anxiety sensitivity, and math anxiety was selected by the main criteria considered, including adjusted R^2 , AIC, BIC, PRESS, LOOCV RMSE, and 10-fold cross-validation RMSE. This suggests that asking-for-help anxiety is not best explained by math anxiety alone; instead, it appears to involve a combination of lower self-efficacy, higher general anxiety sensitivity, and higher math anxiety. This result has practical implications: support for students with statistics anxiety may need to address both quantitative confidence and broader anxiety-related concerns, rather than focusing only on mathematical skill or prior preparation.

Several limitations should be noted. First, the data are observational and cross-sectional, so the regression models describe associations and prediction patterns, not causal effects. Second, the outcomes and predictors are composite scores based on bounded Likert-type survey responses. Although composite scores were treated as approximately continuous, the bounded scale structure contributed to visible banding in residual plots and some significant normality tests. Third, several models showed evidence of heteroscedasticity or mild tail deviations. Robust standard error checks generally preserved the substantive conclusions, but these assumption concerns should still be considered. Fourth, the study sample consisted of university students from a specific dataset, so the findings may not generalize to all student populations or instructional contexts without further validation.

Future work could improve on this analysis in several ways. First, the selected models could be tested on an independent student sample to evaluate external generalizability. Second, longitudinal data would allow researchers to examine whether math anxiety, self-efficacy, or anxiety sensitivity predict later changes

in statistics anxiety over time. Third, future models could include additional instructional variables, such as course format, instructor support, grading structure, or previous grades, to better explain why some students experience higher statistics anxiety. Finally, intervention studies could test whether improving self-efficacy, reducing math anxiety, or providing structured help-seeking support leads to measurable reductions in statistics anxiety.

Overall, the analyses show that statistics anxiety is multidimensional and cannot be fully explained by a single academic background variable. Faculty differences exist but are modest; math anxiety is a central predictor of statistics test anxiety; perfectionism-related concern over mistakes has a positive linear association with interpretation anxiety; prior statistics experience moderates the math anxiety–statistics anxiety relationship; and asking-for-help anxiety is best predicted by a combination of self-efficacy, anxiety sensitivity, and math anxiety. Together, these results support the value of using regression modeling to identify both general and dimension-specific predictors of statistics anxiety. This study also demonstrates how applied regression methods can be used to address educationally meaningful problems and offers valuable empirical evidence for both statistics education and student support practices.

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